

Building 106: Chilled Water System Sequence of Operation

Chiller Sequencing:

The entire chiller plant (which includes the chillers, primary chilled water pumps, secondary chilled water pumps, cooling towers and condenser water system) will operate via a single on off Boolean variable which will be manually set by the operator.

Either the Plate Heat Exchanger or the minimum required number of Chillers shall be run to accommodate the Secondary Chilled Water System load. Chillers shall be started or stopped in a pre defined sequence to meet load requirements.

Stage 1: Chiller #4 shall be the first chiller and shall always run (either in Mechanical Cooling mode or, Plate Heat Exchanger mode) unless the entire CHW plant is commanded off by the operator.

Stage 2: Chiller #3 shall operate as the second chiller.

Stage 3: Chiller #2 shall operate as the third chiller.

Stage 4b: Chiller #6 shall operate as the fourth chiller, **Stage 4a:** and if able drop Chiller #2.

Stage 5b: Chiller #7 shall operate as the fifth chiller, **Stage 5a:** and if able drop Chiller #2.

Chillers 2, 6, and 7 are fixed flow chillers.

Chillers 3 and 4 are variable flow chillers.

Chiller 3 and 4's associated CHW Pump VFD's shall be controlled to match the total primary chilled water loop flow to the total secondary chilled water loop flow in an attempt to operate Chillers 2, 6, and 7 at full load.

If chiller 3 or 4 is the only chiller running, then that chiller's chilled water pump VFD will be modulated between its minimum and maximum flow rate to match the sum of flow meters 1, 2, and 3 on the secondary chilled water system plus a constant (adj.) representing building 103's estimated flow.

CH #3 or #4 flow = FM1 + FM2 + FM3 + Bldg 103

If chillers 3 and 4 are both running, each will control their respective chilled water pumps to maintain half of the sum of flow meters 1, 2, and 3 on the secondary chilled water system plus a constant (adj.) representing building 103's estimated flow.

CH #3 and #4 flow = [FM1 + FM2 + FM3 + Bldg 103] / 2

When additional chillers are running, chillers 3 and 4 each will control their respective chilled water pumps to maintain half of the sum of flow meters 1, 2, and 3 on the secondary chilled water system plus a constant (adj.) representing building 103's estimated flow minus the constant primary flow the additional running chillers.

$$\text{CH\#3 flow} = [\text{FM1} + \text{FM2} + \text{FM3} + \text{Bldg 103} - \text{CH2 flow} - \text{CH6 flow} - \text{CH7 flow}] / 2$$

$$\text{CH\#3 flow} = \text{CH\#4 flow}$$

If chillers 2, 6, & 7 are all off, the commanded CHW temperature setpoint for chillers 3 and 4 will equal the Secondary Chilled Water setpoint. Otherwise the commanded CHW setpoint will be a constant 42° F (adj) to match the manual setpoints on chillers 2, 6, & 7.

(There is a potential for an over pressure condition if too many Liebert computer room valves close in the system, as all AHU chilled water valves will already be closed).

If any Chiller in the above sequence fails, when commanded to start and run, (off for maintenance, in an alarm condition, or mechanical failure), after a 5 min timer (Adj), the next chiller in sequence shall be commanded to start and run.

If there are no additional Chillers, and plant load cannot be maintained, an alarm shall be generated.

Chiller Sequencing Increment:

The initial Secondary Chilled Water System set point shall be 48°F (adj.) and be reset down to 42°F (adj.) based on cooling demand signaled from all large Air Handling Units served by the Secondary Chilled Water System.

Each large Air Handling Units shall issue a cooling demand signal of "1" if the cooling PID is greater than 90% and a demand of "0" if less than 90%.

All large Air Handling Units cooling demand signals shall be summed.

If the Sum is "0" for 5 min (Adj), the Secondary Chilled Water System set point shall be reset up 0.250 degrees F (Adj).

If the Sum is "1 or greater" for 5 min (Adj), the Secondary Chilled Water System set point shall be reset down 0.250 degrees F (Adj).

If Secondary Chilled Water System Supply Temp is greater than the setpoint + 1°F (adj) for 30 minutes (Adj) increment to the next Chiller in sequence.

Chiller Sequencing Decrement:

If the sum, of the currently running chiller capacities in tons, minus the capacity in tons, of the last commanded running chiller in sequence, is greater than the calculated load of the Secondary Chilled Water System load in Tons plus 100, for 10 minutes (Adj), then decrement the number of Chillers called to run.

Chiller Control Sequences:

Chiller #2 Sequence

The Chiller system will occupy one of two states:

1. "RUN"
2. "STOP"

A Boolean point will be defined to set the run or stop state of the Chiller and its associated CHW pump, CW pump and cooling tower.

This Boolean point will be under automatic control, or set to manual override by the operator.

When the Boolean command point is set to run, after a 5 second delay the CHW pump will start.

After an additional 15 second delay the CW pump will start.

Once the status of both the CW and CHW pumps are true for 15 additional seconds, the Chiller Stop Start will be set true.

When the CW pump status is true, the Cooling tower will be enabled based on CW PID.

The lead cooling tower fan VFD shall start when the PID exceeds 20% and lag cooling tower fan VFD shall start when the PID exceeds 30%. Both Fans receive the same speed command and when both are enabled they shall modulate together maintain a condenser water temperature setpoint. Lead/Lag is swapped every Sunday at 10PM.

As outside air conditions permit, the condenser water setpoint will be reset down to a minimum of 65°F based on chilled water supply temperature as follows:

Condenser Water Setpoint = Chilled Water Supply Temperature + 20°F (adj), with a maximum of 85° F.

The Bypass valve will modulate based on the following CW temperature reset schedule:

Below 60°F the bypass valve will be open.

Above 65°F the bypass valve will be closed.

If in a running state, the Boolean command point is set to stop:

1. The Chiller Start-Stop will stop
2. The CW pump will stop after a 1 minute delay (adjustable)
3. The CHW pump will stop after a 2 minute delay (adjustable)

Chiller #3 Sequence

The Chiller system will occupy one of two states:

1. "CHLR" (mechanical cooling) mode,
2. "OFF" mode.

A Boolean point will be defined to set the run or stop state of the Chiller and its associated CHW pump, CW pump, and cooling towers.

This Boolean point will be under automatic control, or set to manual override by the operator.

When indexing from "OFF" mode to "CHLR" mode:

1. The chilled water pump will be started and ramped up to 100% of its flow rate over a two minute period.
2. The condenser water pump will be started.
3. Once the chilled water and condenser water pumps have proven that they are running the chiller will be enabled to run.
4. If the outside air temperature has been above 65°F for 24 hours, and the condenser water supply temperature is also above 65°F, startup is complete. The condenser water bypass valve will be commanded closed. The condenser water pump VFD, lead cooling tower fan VFD, then lag cooling tower fan VFD will be modulated to maintain a 35-psi (adj.) differential refrigerant pressure setpoint based on the highest reading between the lead and lag chillers. 0-50% of the PID is reset to 0-100% speed command for the CW pump. 50-100% of the PID is re-scaled to 17-100% for tower speed control. The lead cooling tower fan VFD shall start when the re-scaled value exceeds 50% and the lag cooling tower shall start when the re-scaled value exceeds 55%. Both fans receive the same speed command and when enabled they modulate together. Lead/Lag is swapped every Sunday at 10PM.
5. If the outside air temperature has not been above 65°F for 24 hours, the controller will enter "Condenser System Warm-up Mode". The condenser water bypass valve will be commanded to 100% open (full bypass of the cooling tower). The condenser water pump will run at minimum speed. The cooling tower fans will remain off. The running chiller will warm up the fully bypassed condenser water loop. Once the condenser water supply temperature is above 65°F, the condenser water pump speed will increase to 100% and the condenser water bypass valve will be limited to no more than 10%. A PID control loop will modulate the condenser water bypass valve from 10% towards 0% to maintain a 65°F condenser water supply temperature setpoint. Once the condenser water bypass valve is fully closed, startup is complete. The condenser water bypass valve will be commanded closed. The condenser water pump VFD, lead cooling tower fan VFD, then lag cooling tower fan VFD (in order) will be modulated to maintain a 35-psi (adj.) differential refrigerant pressure setpoint based on the highest reading between the lead and lag chillers.

After the above startup sequence is complete the chilled water pump VFD speed will modulate under control of the chiller manager program.

When indexing from "CHLR" mode to "OFF" mode:

1. The chillers will be commanded off.
2. After the chillers are off, the chilled water pump will run for a minimum of three minute (adj.) before being commanded off.
3. After the chillers are off, the condenser water pump will run for a minimum of two minutes (adj.) before being commanded off.
4. The cooling tower fans will stop.

Chiller #4 Sequence

The Chiller and heat exchanger system will occupy one of three states.

1. "HEX" (heat exchanger free cooling) mode,
2. "CHLR" (mechanical cooling) mode,
3. "OFF" mode.

When chiller 4 and its associated heat exchanger are in the "OFF" mode, the chiller isolation valves will remain open, and the heat exchanger isolation valves will remain closed. The lead chiller (chiller A), the lag chiller (chiller B), the CHW pump, the CW pump, and cooling tower fan VFD's will be off.

When commanded to run, OA temperature will be used to determine whether to run in "HEX" mode, or "CHLR" mode.

If the outside air temperature is below 46°F (adj.) for a minimum time of sixty (60) minutes (adj.) the unit will index to "HEX" mode when commanded to run.

If the outside air temperature is above 49°F (adj.) for a minimum time of sixty (60) minutes (adj.), the unit will index to "CHLR" mode when commanded to run.

When indexing from "OFF" mode to "CHLR" mode:

1. The chilled water isolation valves on the chiller will remain open and the chilled water isolation valves on the heat exchanger will remain closed.
2. The chilled water pump will be started and ramped up to 100% of its flow rate over a two minute period.
3. The condenser water pump will be started.
4. Once the chilled water and condenser water pumps have proven that they are running the chiller will be enabled to run.
5. If the outside air temperature has been above 65°F for 24 hours, and the condenser water supply temperature is also above 65°F, startup is complete. The condenser water bypass valve will be commanded closed. The condenser water pump VFD, lead cooling tower fan VFD, then lag cooling tower fan VFD will be modulated to maintain a 35-psi (adj.) differential refrigerant pressure setpoint based on the highest reading between the lead and lag chillers. The lead cooling tower fan VFD shall start when the PID exceeds 50% and lag cooling tower fan VFD shall start when the PID exceeds 55%. Both Fans receive the same speed command and when both are enabled they shall modulate together maintain a condenser water temperature setpoint. Lead/Lag is swapped every Sunday at 10PM.
6. If the outside air temperature has not been above 65°F for 24 hours, the controller will enter "Condenser System Warm-up Mode". The condenser water bypass valve will be commanded to 100% open (full bypass of the cooling tower). The condenser water pump will run at minimum speed. The cooling tower fans will remain off. The running chiller will warm up the fully bypassed condenser water loop. Once the condenser water supply temperature is above 65°F, the condenser water pump speed will increase to 100% and the condenser water bypass valve will be limited to no more than 10%. A PID control loop will modulate the condenser water bypass valve from 10% towards 0% to maintain a 65°F condenser water supply temperature setpoint. Once the condenser water bypass valve is fully closed, startup is complete. The condenser water bypass valve will be commanded closed. The condenser water pump VFD, lead cooling tower fan VFD, then lag cooling tower fan VFD (in order) will be modulated to

maintain a 35-psi (adj.) differential refrigerant pressure setpoint based on the highest reading between the lead and lag chillers.

After the above startup sequence is complete the chilled water pump VFD speed will modulate under control of the chiller manager program.

When indexing from "OFF" mode to "HEX" mode:

1. The chiller isolation valves will close and the heat exchanger isolation valves will open.
2. The cooling tower bypass valve for chiller 4 will be positioned to route all condenser water through the tower.
3. The chilled water pump will be started and ramped up to 100% of its flow rate over a two minute period.
4. The condenser water pump VFD will modulate to maintain a primary chilled water setpoint of 48°F (adj.).
5. The lead cooling tower fan VFD and lag cooling tower fan VFD (in order) will modulate to maintain a condenser water temperature setpoint reset as follows:

<u>OAT</u>	<u>CW Setpoint</u>
40	45
43	48

After the above startup sequence is complete the chilled water pump VFD speed will modulate under control of the chiller manager program.

When indexing from "CHLR" mode to "OFF" mode:

- a. The chillers will be commanded off.
- b. After the chillers are off, the chilled water pump will run for a minimum of three minute (adj.) before being commanded off.
- c. After the chillers are off, the condenser water pump will run for a minimum of two minutes (adj.) before being commanded off.
- d. The cooling tower fans will stop.

When indexing from "HEX" mode to "OFF" mode:

1. The isolation valves on the chiller will open and the heat exchanger isolation valves will close.
2. The chilled water pump will stop.
3. The condenser water pump will stop.
4. The cooling tower fans will stop.

When indexing from "HEX" mode to "CHLR" mode:

1. The chilled water isolation valves on the chiller will open and the chilled water isolation valves on the heat exchanger will close.
2. The condenser water pump will remain running but at minimum speed.
3. The chilled water pump will remain running.
4. If the chilled water and condenser water pumps have proven that they are running the chiller will be enabled to run.
5. "Condenser System Warm-up Mode" will initiate. The condenser water bypass valve will be commanded to 100% open (full bypass of the cooling tower). The condenser water pump will run at minimum speed. The cooling tower fans will remain off. The running chiller will warm up the fully bypassed condenser water

loop. Once the condenser water supply temperature is above 65°F, the condenser water pump speed will increase to 100% and the condenser water bypass valve will be limited to no more than 10%. A PID control loop will modulate the condenser water bypass valve from 10% towards 0% to maintain a 65°F condenser water supply temperature setpoint. Once the condenser water bypass valve is fully closed, startup is complete. The condenser water bypass valve will be commanded closed. The condenser water pump VFD, lead cooling tower fan VFD, then lag cooling tower fan VFD (in order) will be modulated to maintain a 35-psi (adj.) differential refrigerant pressure setpoint based on the highest reading between the lead and lag chillers.

After the above startup sequence is complete the chilled water pump VFD speed will modulate under control of the chiller manager program.

When indexing from "CHLR" mode to "HEX" mode:

1. The chillers will be commanded off.
2. The chilled and condenser water pumps will remain running.
3. A sequence hold will occur until the status of chillers 4a and 4b are off.
4. The chiller isolation valves will close and the heat exchanger isolation valves will open.
5. The cooling tower bypass valve for chiller 4 will be positioned to route all condenser water through the tower.
6. The condenser water pump VFD for chiller 4 will ramp up and modulate to maintain a primary chilled water setpoint of 48°F (adj.).
7. The lead cooling tower fan VFD and lag cooling tower fan VFD (in order) will modulate to maintain a condenser water temperature setpoint reset as follows:

<u>OAT</u>	<u>CW Setpoint</u>
40	45
43	48

After the above startup sequence is complete the chilled water pump VFD speed will modulate under control of the chiller manager program.

Cooling Tower Defrost Mode:

When OA Temp is < 33 deg F (adj), AND chiller plant #4 is in "HEX", either by automatic or by manual setting, than "cooling tower #4 defrost" shall be enabled. When OA temp rises to more than 1 deg F (adj) above the cooling tower #4 defrost enable setpoint (the deadband), cooling tower #4 defrost shall be disabled.

Cooling tower #4 defrost shall disable the CT#4 fans, one at a time, according to the following adjustable reset schedule:

OAT (deg F)	Time Between Defrosts	Time each fan is in Defrost shutdown
Defrost Enable temp	6 Hr, 0 Min	0 Hr, 15 Min
	02 Hr, 0 Min	0 Hr, 30 Min

Chiller 6 & 7 Sequence

The Chiller system will occupy one of two states:

1. "RUN"
2. "STOP"

A Boolean point will be defined to set the run or stop state of the Chiller and its associated CHW pump, CW pump and cooling tower.

This Boolean point will be under automatic control, or set to manual override by the operator.

When the Boolean command point is set to run, after a 5 second delay the CHW pump will start.

After an additional 15 second delay the CW pump will start.

Once the status of both the CW and CHW pumps are true for 15 additional seconds, the Chiller Stop Start will be set true.

When the CW pump status is true, the Cooling tower will be enabled based on CW PID.

The lead cooling tower fan VFD shall start when the PID exceeds 20% and lag cooling tower fan VFD shall start when the PID exceeds 30%. Both Fans receive the same speed command and when both are enabled they shall modulate together maintain a condenser water temperature setpoint. Lead/Lag is swapped every Sunday at 10PM.

As outside air conditions permit, the condenser water setpoint will be reset down to a minimum of 65°F based on chilled water supply temperature as follows:

Condenser Water Setpoint = Chilled Water Supply Temperature + 20°F (adj), with a maximum of 85° F.

If in a running state, the Boolean command point is set to stop:

1. The Chiller Start-Stop will stop
2. The CW pump will stop after a 1 minute delay (adj.)
3. The CHW pump will stop after a 2 minute delay (adj.)

Secondary Chilled Water Pumps #1, #2 and #3 Sequence:

The Secondary Chilled Water Pumps (SCHWP) shall be indexed to run, as needed, any time the chiller plant is manually started.

The SCHWPs shall maintain equal run times during the year. They will be alternated every Sunday:

- P1, P2, P3
- P2, P3, P1
- P3, P1, P2

When the lead pump is indexed to run, it shall ramp up to maintain the differential pressure setpoint.

The differential pressure setpoint will be reset between 5 and 12 psi (adj), (as read at the differential pressure sensor located in the building 207 vault) based on an OA enthalpy of 22 btu/lb to 40 btu/lb (adj).

When the differential pressure PID is greater than 60% output, the second SCHWP will be indexed to run and both pumps will be controlled at the same speed to maintain the differential pressure setpoint.

When the differential pressure PID is greater than 60% again, the third SCHWP will be indexed to run and all three pumps will be controlled at the same speed to maintain the differential pressure setpoint.

When the differential pressure PID is less than 20%, the third SCHWP started will be indexed to stop and the remaining two SCHWPs will be controlled at the same speed to maintain the differential pressure setpoint.

When the differential pressure PID is less than 20% output again, the second SCHWP will be indexed to stop and the lead SCHWP will be controlled to maintain the differential pressure setpoint.

An alarm will be activated if the differential pressure is above or below setpoint by more than 0.4 psi (adj.) for more than 2 min, 30 seconds (adj.)

An alarm will be activated if any of the three pumps speed feedback is 5% (adj.) above or below commanded speed for more than 30 seconds.

Additional Notes for all chillers:

While any chiller is running, if either the associated CHW or CW pump statuses drop out, the chiller stop/start will go off.

The Lead and Lag cooling towers assignment will swap each Sunday at 10PM.

An alarm will be provided if the CW supply temp is above or below the set point by 5 degrees (adj.) for 15 minutes.

An alarm will be provided if the CW pump or CHW pump does not match the commanded state within 1 minute.

An alarm will be provided if the Chiller status (if provided) does not match the commanded state within 3 minute.

Upon a Chiller failure, that Chiller will be locked-out and require an operator reset on graphic. Lock-out shall disable pumps after their time delays.

END OF SEQUENCE